I. Optimization Problem

Modeling of an Optimization Problem

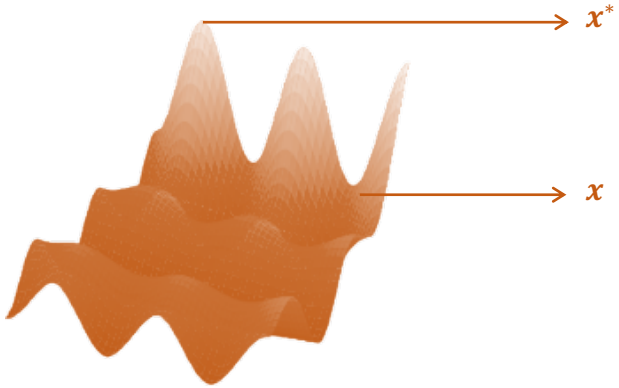
$$\min_{x \in \mathbb{R}} f(x) \quad \Longrightarrow \quad f(x^*) \leq f(x), \forall x \in X$$

X: space of feasible (potential) solutions.

$x = (x_1, x_2, \dots, x_D)$: decision variable vector.

$f(x)$: objective function to be minimized (or maximized).

x^*: optimal solution or near-optimal solution.



Classification of Optimization Problems:

Discrete / Continuous

Single-objective / Multi-objective

Unconstrained / Constrained

Solving Optimization Problems:

Exact methods

Approximate methods (Metaheuristics)

Metaheuristics:

Intensification (Exploitation)

Diversification (Exploration)

II. Standard Continuous Optimization Benchmark Problems in Metaheuristics

Main benchmark functions:

Unimodal (UM) functions: F1–F7

Multimodal (MM) functions: F8–F23

Examples:

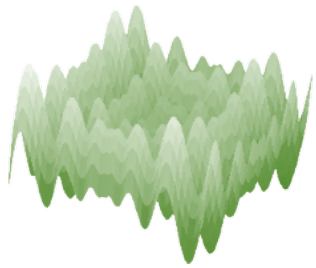
Unimodal benchmark functions

Function	Description	Dimensions (D)	Range (x_i)	f_{min}
F1	$f(x) = \sum_{i=1}^{D} x_i^2$	30, 100, 500, 1000	[-100, +100]	0
F2	$f(x) = \sum_{i=1}^{D} \lvert x_i \rvert + \prod_{i=1}^{D} \lvert x_i \rvert$	30, 100, 500, 1000	[-10, +10]	0
F5	$f(x) = \sum_{i=1}^{D-1} [100(x_i^2 - x_{i+1})^2 + (1 - x_i)^2]$	30, 100, 500, 1000	[-30, +30]	0
F7	$f(x) = \sum_{i=1}^{D} i x_i^4 + \mathrm{rand}(0, 1)$	30, 100, 500, 1000	[-128, +128]	0

Multimodal benchmark functions

Function	Description	Dimensions (D)	Range (x_i)	f_{min}
F8	$f(x) = \sum_{i=1}^{D} \left(-x_i \sin\left(\sqrt{\lvert x_i \rvert}\right)\right)$	30, 100, 500, 1000	[-500, +500]	-418.9829 × D
F9	$f(x) = \sum_{i=1}^{D} [x_i^2 - 10\cos(2\pi x_i) + 10]$	30, 100, 500, 1000	[-5.12, +5.12]	0
F11	$f(x) = 1 + \frac{1}{4000} \sum_{i=1}^{D} x_i^2 - \prod_{i=1}^{D} \cos\left(\frac{x_i}{\sqrt{i}}\right)$	30, 100, 500, 1000	[-600, +600]	0

Role

Function	Testing	Number of Global Optima	Number of Local Minima	Role	Example	Visualization
Unimodal	Exploitation	One	None	Evaluate the exploitation capability and convergence behavior of the algorithm	F4	
Multimodal	Exploration	One or multiple	Large	Evaluate exploration ability and avoidance of local minima	F13	

Exercise :

Implement the benchmark functions and test them using a randomly generated candidate solution based on the domain of each function.

TP - Metaheuristics

Part 1 \ Optimization Problem Initialization

Standard Continuous Optimization Benchmark Problems in Metaheuristics

	Dimension (D)	Range		
Solution:	30	-100	+100	Generate solution

Candidate solution example

Fitness

Function: F1-UM

$$f(x) = \sum_{i=1}^{D} x_i^2$$

Evaluate solution

Candidate solutions to be evaluated:

Function	Solution
F1	-27.81 | -71.96 | -47.13 | 54.63 | -86.58 | -96.77 | 63.39 | 75.60 | -39.94 | -45.13 | 90.77 | 70.68 | 36.61 | 18.50 | -46.11 | -91.04 | -34.74 | 94.34 | 61.98 | -77.94 | -78.75 | -3.11 | -2.81 | 80.69 | -76.95 | 43.46 | 3.65 | -26.73 | 49.26 | 0.72
F2	9.79 | -1.60 | 6.62 | 6.07 | 8.16 | -1.29 | 3.97 | 7.41 | 1.13 | -7.32 | 7.15 | 3.59 | 5.92 | -9.07 | 2.34 | -2.47 | -7.22 | 4.24 | -7.75 | -9.37 | 7.08 | 5.02 | -2.03 | -4.33 | 1.50 | -2.29 | 7.32 | 9.72 | -9.67 | -2.52
F5	-23.85 | -19.42 | 6.20 | -9.92 | -1.11 | 6.19 | -3.25 | 10.30 | 22.39 | -10.27 | 8.92 | 28.76 | -10.92 | 28.92 | 12.16 | -23.71 | 13.74 | -12.67 | -23.86 | 19.71 | -25.63 | 28.86 | -28.97 | 11.99 | 6.37 | -8.29 | -6.86 | 1.50 | -25.89 | 19.02
F7	15.30 | -124.59 | 15.30 | 76.61 | -24.73 | 60.58 | -37.78 | 89.80 | -77.46 | 71.79 | 1.68 | 54.56 | -9.37 | -48.23 | 10.54 | -46.11 | -94.84 | 9.99 | -10.71 | 19.30 | -46.78 | 20.22 | -109.43 | -66.11 | 103.04 | 124.89 | -33.52 | 43.97 | -52.42 | -71.94
F8	294.94 | 93.76 | 77.03 | -44.49 | 338.08 | 252.79 | 318.38 | 54.94 | 428.00 | 466.38 | 306.97 | 344.46 | 469.53 | 251.11 | -198.22 | 182.60 | 362.80 | 322.47 | 377.28 | 237.63 | 131.96 | -58.09 | -240.77 | -484.17 | -464.34 | -152.18 | -38.75 | 369.36 | -135.30 | -249.50
F9	-1.20 | -1.63 | 2.42 | -4.95 | -2.28 | 1.16 | 0.02 | 0.71 | 0.80 | 4.00 | 4.92 | -1.61 | -0.77 | 2.36 | 4.36 | -5.09 | -1.42 | 0.21 | -1.96 | 3.47 | 3.08 | -3.75 | 3.71 | 3.97 | 3.80 | -3.91 | -0.09 | 4.51 | -1.06 | 0.68
F11	377.01 | -369.41 | 81.54 | -546.71 | -60.41 | -338.54 | -561.90 | -268.59 | -21.93 | 118.09 | 150.80 | -507.26 | 25.70 | -540.82 | 146.39 | 243.18 | -107.89 | 242.99 | -281.18 | 195.16 | 301.57 | 78.46 | -447.01 | -492.81 | -488.54 | -24.41 | -22.75 | -324.18 | 437.47 | -595.42